

Analysis I

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Preface

You have already learned how to do calculus and somewhat understand why it works. So, you might wonder, “*Why study calculus again? Why do we need to justify it formally?*”

It’s a fair question. We already have an intuitive idea of how *infinitesimal calculus* works: the derivative is defined as the slope of a secant line on a curve when the points on the curve come “infinitesimally close” to each other, so close that they almost overlap but are still two distinct points.

But have you ever wondered what “infinitesimally close” or “infinitesimally small” quantities even mean? You might say, “It means that the quantity is *very small*, almost close to zero but not zero *itself*, such as 10^{-40} ”.

But examples like that describe *concrete* small quantities, it is a *concrete* value of “how small”. No matter how small a quantity you pick, “infinitesimally small” quantities are supposed to be always smaller than them.

So, infinitesimals seems like this vague, intuitive idea which we can’t pin down precisely. You may wonder, “How do you *precisely* pin down something at the first place?”

The idea of “closeness” is formalised in a topological concept called *neighbourhoods*, which essentially says that two points are “close” if they both lie in a small interval (or ball) around a point. Using this idea of neighbourhoods, we can define limits, continuity, derivatives and integrals rigorously.

See, in mathematics, we try to prove statements from first principles formally, and visual demonstrations or informal intuitive explanations are not accepted as proofs.

This topic was discussed in more detail from a philosophical perspective in the notes on *Mathematical Foundations* and the crux of it is this: our informal intuitions are based on assumptions that are not explicitly pinned down, in formal mathematics we try to pin down every assumption explicitly.

So, the reason for studying analysis is not just to understand why calculus works and when our rules apply, it is to understand what *formally proving* something even means and why we do it.

“*Okay, so you’re saying that our informal intuitions hide some assumptions but what exactly are those assumptions?*”

The assumptions for calculus actually start from very basic concepts in maths, from numbers and arithmetic itself. Because we deal with real numbers in analysis we need to discuss what real numbers mean and construct it axiomatically from primitive axioms. The need for

this is that there is some ambiguity in our understanding of numbers, for example, do you know if $0.999\dots$ *exactly* equals 1 or is this just an approximation? And does infinite sum of a geometric series *exactly* equal to its limit? We say that it “approaches this value”, but why shouldn’t it just keep on increasing forever, even if by extremely tiny amounts?

The reason why I think analysis is so hard even for the best students is not just because it’s the first time dealing with abstraction and proofs, but also that students are not given proper motivation. I will not try to rush through the concepts. I’ll show you that analysis is not “Calculus with Proofs” but “Justification of Calculus”, a subtle but important distinction.

In the first volume, we’ll cover the real, complex and extended fields, basics of general topology, sequences and series, limits and continuity, differential and integral calculus of functions of single real variables and the two fundamental theorems of calculus. We also cover additional topics on sequence and series of functions, Fourier analysis and improper integral.

Although we will discuss some basic concept of functions of complex variables and how functions form vector spaces as well, a detailed coverage will not be done here. You can refer to the notes on *Complex Analysis* and *Functional Analysis* for a detailed coverage.

The **prerequisites** for this course are working knowledge of logic, sets and proof techniques, you can refer to my notes on *Mathematical Foundations* for those topics. Some knowledge of groups, rings, fields and vector spaces would also be helpful but that can be covered alongside this course. Apart from these, only high school mathematics is expected from the reader.

These notes were prepared between October 2025 and (tentative) (**Last update: February 2, 2026**).

References

These lecture notes closely follow the textbooks by Amann and Escher, however these two lecture notes are also useful although, they are written in German.

- Analysis I - WiSe 2016/17, by Franz Merkl, Fakultät für Mathematik, Informatik und Statistik, LMU München.
- Analysis einer Veränderlichen - WiSe 2013/14, by Lars Diening, Fakultät für Mathematik, Informatik und Statistik, LMU München.

These books will serve as great references. There’s a mix of book English and German texts, although you can find English translations for some of them.

- Herbert Amann, Joachim Escher, Analysis I, Dritte Auflage
- Otto Forster, Florian Lindemann, Analysis 1, 13. Auflage
- Walter Rudin, Principles of Mathematical Analysis, 3rd. Edition
- K. Königsberger, Analysis 1

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